

Week 3 Weekly Report

1. Work Completed

During Week 3, the following tasks were completed:

- Reviewed the Black-Scholes-Merton framework for European call and put option pricing.

- Implemented reusable functions for the standard normal distribution, model-input validation, and BSM call and put valuation.

- Implemented the analytical simple chooser option model and the theoretically correct choice boundary at the six-month decision date.

- Configured the contract with a strike price of \$150, a choice time of 0.5 year, and a final maturity of one year.

- Separated the paper replication parameters from the updated Week 2 market-data baseline.

- Linked the Week 3 model to the processed Week 2 JPM close, Treasury rate, and 20-day annualized historical volatility.

- Compared the recalculated choices and payoffs with the ten observations reported in the paper's Table 3.

- Validated the analytical chooser price using a reproducible two-period Monte Carlo simulation with 250,000 paths.

- Exported parameter, pricing, paper-comparison, Monte Carlo, and validation results for independent review.

2. Deliverables

Completed deliverables include:

- Week 3 BSM Chooser Option Model Notebook (Week3.ipynb)

- Week 3 Model Validation Notebook (Week3_model_validation.ipynb)

- Parameter Configuration File (parameter_config.json)

- Pricing and Validation Result Files (model_results)

- Week 3 Learning Report

3. Progress Summary

All planned Week 3 objectives have been completed.

A functional BSM-based simple chooser option model has been established. The implementation reproduces the paper's parameter configuration, connects to the processed Week 2 dataset, and provides both analytical and simulation-based validation. This model now serves as the baseline for the Week 4 performance evaluation.

4. Experimental Results

Table 1. Parameter Configuration

Parameter	Paper	Week 2 Baseline	Source or Treatment
Spot price S_0	\$156.7000	\$231.0776	Paper Table 2 / latest Week 2 close
Strike K	\$150.0000	\$150.0000	Required project setting
Risk-free rate r	0.15%	4.55%	Paper Table 2 / Week 2 Treasury proxy
Dividend yield q	2.33%	2.33%	Paper value retained; not collected in Week 2
Volatility σ	28.20%	19.0625%	Paper Table 2 / Week 2 20-day annualized volatility
Choice time T_1	0.5 year	0.5 year	Six-month decision date
Maturity T_2	1 year	1 year	Required project setting

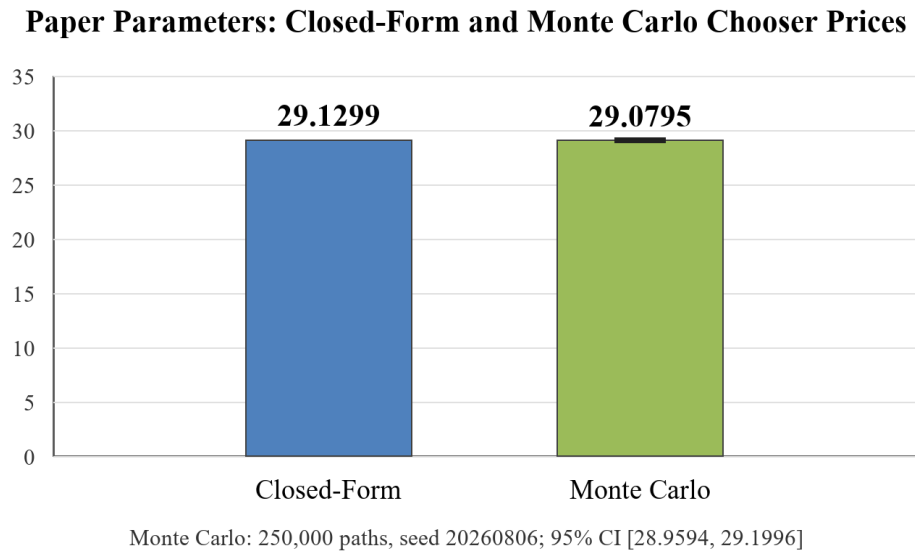
The paper configuration is used for replication, while the updated baseline demonstrates the data connection from Week 2. The Week 2 Treasury value is converted from 4.55 to 0.0455 before entering the BSM formulas.

Table 2. Analytical Pricing Results

Configuration	European Call	European Put	Chooser Option
Paper replication	\$18.6890	\$15.3731	\$29.1299
Week 2 updated baseline	\$82.5255	\$0.0977	\$82.5278

Under the paper parameters, the chooser option is more valuable than either standalone European option because it preserves the right to select the more valuable direction at the choice date. The updated baseline produces a much larger call-dominated value because the latest JPM close is substantially above the fixed \$150 strike.

Figure 1. Closed-Form and Monte Carlo Chooser Prices



The analytical price of \$29.1299 lies inside the Monte Carlo 95% confidence interval of \$28.9594 to \$29.1996. The absolute difference between the two estimates is approximately \$0.0504.

Table 3. Initial Validation Results

Validation Test	Observed Result	Status
Put-call parity error	0.000000	Pass
Chooser value is at least the call value	\$10.4409 margin	Pass
Chooser value is at least the put value	\$13.7569 margin	Pass
Chooser value does not exceed call plus put	\$4.9321 margin	Pass
Paper Table 3 choice comparison	10 of 10 matched	Pass
Paper Table 3 payoff comparison	Maximum error approximately 0	Pass
Monte Carlo vs. closed form	0.823 standard errors	Pass

The paper does not disclose the random seed or random shocks used for Table 3. Therefore, the published payoff calculations can be reproduced exactly, but the same simulated stock-price paths cannot be regenerated from the available information.

5. Issues Encountered

The paper's simulation is not fully reproducible because its random seed and random shocks are not reported. A documented fixed seed was therefore used for the independent Monte Carlo comparison.

The Week 2 dataset does not contain JPM dividend yield. The paper's 2.33% dividend yield was retained for the updated baseline and clearly identified in the configuration file.

The Week 2 risk-free-rate proxy is the 10-year Treasury yield, while the chooser option has a one-year maturity. This maturity mismatch is a limitation of the current baseline.

The fixed \$150 strike is far below the latest Week 2 JPM close, which makes the updated baseline strongly call-dominated. This is an expected consequence of retaining the paper's strike rather than a model error.

6. Plan for Week 4

Obtain suitable actual option transaction or benchmark prices for baseline evaluation.

Calculate MAE and RMSE between BSM model predictions and observed option prices.

Analyze model limitations during high-volatility periods and document the effect of the current data and parameter assumptions.

Establish a clearly documented BSM performance baseline for later comparison with machine-learning models.